

Operations with Complex Numbers

Algebra II Worksheet · Grade 9–11

Name: _____

Date: _____

Learning Objectives

- Simplify square roots of negative numbers using the imaginary unit i
- Write complex numbers in standard form $a + bi$
- Add, subtract, and identify parts of complex numbers

Problems

1. Simplify the square root below using the imaginary unit i :

$$\sqrt{-1}$$

2. Simplify the expression below:

$$\sqrt{-49}$$

3. Simplify the expression below:

$$\sqrt{-5}$$

4. Simplify the expression below:

$$\sqrt{-48}$$

5. Identify the real part and imaginary part of the complex number below:

$$7 + 5i$$

6. Write the complex number in standard form $a + bi$:

$$2i + 6$$



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Complex Numbers — Operations Worksheet

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7. Simplify the expression and write it in standard form:

$$2 - \sqrt{-9}$$

8. Add the two complex numbers below and write in standard form:

$$(8 + 6i) + (3 + 2i)$$

9. Subtract the complex numbers below and write in standard form:

$$(10 + 7i) - (4 + 3i)$$

10. Simplify the square root, then add the complex numbers and write the result in standard form:

$$(5 + \sqrt{-16}) + (3 - \sqrt{-36})$$



Operations with Complex Numbers — Answer Key

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Answer Key

1. Answer: i

- By definition, the square root of negative one is i
 - Answer: i
-

2. Answer: $7i$

- Factor: $\sqrt{-1} * \sqrt{49}$
 - $\sqrt{-1} = i$ and $\sqrt{49} = 7$
 - Answer: $7i$
-

3. Answer: $i\sqrt{5}$

- Factor: $\sqrt{-1} * \sqrt{5}$
 - $\sqrt{-1} = i$, and 5 is not a perfect square so it stays as $\sqrt{5}$
 - Answer: $i\sqrt{5}$
-

4. Answer: $4i\sqrt{3}$

- Factor -48 as $-1 * 16 * 3$
 - $\sqrt{-1} = i$, $\sqrt{16} = 4$, $\sqrt{3}$ stays
 - Answer: $4i\sqrt{3}$
-

5. Answer: Real part: 7, Imaginary part: $5i$

- A complex number is written as $a + bi$
 - $a = 7$ is the real part
 - $b = 5$, so the imaginary part is $5i$
-

6. Answer: $6 + 3i$

- Standard form requires the real part first
 - Real part is 6, imaginary part is $3i$
 - Answer: $6 + 3i$
-

7. Answer: $2 - 3i$

- Simplify $\sqrt{-9}$: $\sqrt{-1} * \sqrt{9} = i * 3 = 3i$
 - Substitute back: $2 - 3i$
 - Answer: $2 - 3i$
-

8. Answer: $11 + 8i$

- Combine real parts: $8 + 3 = 11$
 - Combine imaginary parts: $6i + 2i = 8i$
-



- Answer: $11 + 8i$
-

9. Answer: $6 + 4i$

- Distribute the negative: $(10 + 7i) + (-4 - 3i)$
 - Combine real parts: $10 - 4 = 6$
 - Combine imaginary parts: $7i - 3i = 4i$
 - Answer: $6 + 4i$
-

10. Answer: $8 - 2i$

- Simplify $\sqrt{-16} = 4i$ and $\sqrt{-36} = 6i$
 - Rewrite: $(5 + 4i) + (3 - 6i)$
 - Combine real parts: $5 + 3 = 8$
 - Combine imaginary parts: $4i - 6i = -2i$
 - Answer: $8 - 2i$
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